

# External Validation One-Page

MIDAS Section Library

embedded metadata validated

static + contract gate

MIDAS section-library: ok | 182/183 used | 183 templates | source=midas\_parser\_derived |  
implementation/phase1/open\_data/midas/midas\_generator\_33.json  
nightly, release gap, committee dashboard, and this external validation onepage all consume the same validator line

Validator line

MIDAS section-library: ok | 182/183 used | 183 templates | source=midas\_parser\_derived |  
implementation/phase1/open\_data/midas/midas\_generator\_33.json

|                      |                                                         |
|----------------------|---------------------------------------------------------|
| Bundle               | 20260422T140547Z                                        |
| Generated            | 2026-04-22T14:05:48.342988+00:00                        |
| Nightly              | PASS                                                    |
| CI                   | PASS                                                    |
| Static validation    | PASS                                                    |
| Freeze               | PASS                                                    |
| Promotion            | PASS                                                    |
| Promotion reason     | PASS                                                    |
| Promotion hold       | False                                                   |
| Hold review manifest | implementation/phase1/release/hold_review_manifest.json |
| Hold review packet   | implementation/phase1/release/hold_review_packet.md     |

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|                                 |                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------|
| Hold review packet pdf          | implementation/phase1/release/hold_review_packet.pdf                                           |
| Hold review ack                 | implementation/phase1/release/hold_review_ack.json                                             |
| Smoke reason                    | PASS                                                                                           |
| Smoke pass rate                 | 100.00%                                                                                        |
| Smoke trial feasible            | 100.00%                                                                                        |
| Smoke avg trial runtime (s)     | 0.0563                                                                                         |
| Smoke strict recommendation     | candidate_for_strict_enable                                                                    |
| Signed registry                 | PASS                                                                                           |
| Registry signature              | PASS                                                                                           |
| Solver HIP e2e                  | PASS                                                                                           |
| RC benchmark lock               | PASS                                                                                           |
| NDTHA residual gate             | PASS                                                                                           |
| Committee package               | PASS                                                                                           |
| MIDAS section-library validator | MIDAS section-library: ok   182/183 used   183 templates   source=midas_parser_derived   imple |
| MIDAS section-library status    | embedded metadata validated                                                                    |
| Material constitutive           | Material constitutive gate: PASS   concrete_damage=yes(matrix=48/48,max=1.000)   cyclic_deg    |
| Surface interaction benchmark   | Surface interaction benchmark: PASS   ready=7/7   family_matrix=420/420   source_families=10   |
| KDS row provenance export       | MIDAS KDS row provenance export: PASS   combos=6   rows=144   members=12   clauses=6   e       |

# External Validation One-Page (Continued)

|                                   |                                                                                                   |
|-----------------------------------|---------------------------------------------------------------------------------------------------|
| KR ingest                         | KR ingest: PASS   src=15   cls=4   got=10   fp=10   meta=5   rej=0   dup=0   seed=15   topo=5   n |
| Measured benchmark breadth        | Measured benchmark breadth: PASS   baseline=2/51   opensees_delta=6/7   authority_delta=2/6       |
| OpenSees canonical breadth        | OpenSees canonical breadth: PASS   families=6   cases=7   parser_ready=3   origins=designsafe_    |
| KR preview queue                  | KR preview queue: PASS   cand=8   pend=0   state=closed_until_new_public_archive_exact_topo       |
| KR representative batches         | building=kci_concrete_building_design_examples_2e_native_baseline   housing=lh_bucheon_ye         |
| Irregular structure track         | Irregular structure collection gate: PASS   families=20   sources=32   local_ready=15   remote_ca |
| Irregular benchmark execution     | Irregular benchmark execution: CHECK   ready=0   blocked=0   task_count=0   canonical=0   bric    |
| Commercial grade                  | Commercial                                                                                        |
| Deployment model                  | engineer_in_the_loop_accelerated_coverage                                                         |
| Accelerated coverage              | 95-99%                                                                                            |
| Residual holdout                  | 1-5%                                                                                              |
| Estimated time saved              | 90-96%                                                                                            |
| Empirical smoke runtime reduction | 95.66-96.04%                                                                                      |
| Comparable run mode               | current_pipeline_comparable_full_chain_pass                                                       |
| Comparable deployment             | engineer_in_the_loop_accelerated_coverage                                                         |
| Comparable strict smoke           | True                                                                                              |
| Engineer-in-loop ready            | True                                                                                              |
| Estimated time basis              | Empirical estimate derived from nightly design-optimization smoke runtime reduction, scaled by    |

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|                                               |                                                                                                 |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------|
| Time-saving focus                             | Use this engine to automate the dominant, time-consuming 95-99% of repeated analysis, screening |
| Full replacement ready                        | False                                                                                           |
| External benchmark start now                  | True                                                                                            |
| External benchmark full submission ready      | True                                                                                            |
| External benchmark reason                     | PASS_START_NOW_FULL                                                                             |
| External benchmark start mode                 | start_now_full_external_submission                                                              |
| External benchmark scope                      | full_external_submission_package                                                                |
| External benchmark blockers                   | none                                                                                            |
| External benchmark cautions                   | panel_zone_external_validation_only_boundary                                                    |
| External benchmark execution mode             | limited                                                                                         |
| External benchmark execution ready tasks      | 10                                                                                              |
| External benchmark execution blocked tasks    | 0                                                                                               |
| External benchmark execution review-boundary  | 0                                                                                               |
| External benchmark review-boundary resolution | approve_all=PASS_NO_OPEN_DECISION_ITEMS/ready_full=yes; reject_one=PASS_NO_OPEN_D               |
| External benchmark execution status           | execution_complete_no_fail                                                                      |
| External benchmark execution status counts    | planned=0   in_progress=0   completed=10   failed=0   finished=10   completion_ratio=1.000      |
| External benchmark case attestation workflow  | cases=10   manifests=10   templates=0   receipts=10   attested=10   source=manifest=10   statu  |
| PBD response source                           | resolved_report=implementation/phase1/experiments/by_test/nonlinear_ndtha_stress/2026031        |

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|                                          |                                                                                               |
|------------------------------------------|-----------------------------------------------------------------------------------------------|
| Audit review decision batch template     | items=0   status=none   owner=none   priority=none   attested_examples=0   example_preview=   |
| Approve-all readiness preview            | reason=PASS_NO_OPEN_DECISION_ITEMS   ready_full=False   pending=0   open_revision=0           |
| Reject-one readiness preview             | reason=PASS_NO_OPEN_DECISION_ITEMS   ready_full=False   pending=0   open_revision=0   bl      |
| Audit review decision batch runner       | reason=PASS_ZERO_TOUCH_NO_OPEN_DECISION_ITEMS   apply_live=False   live_applied=False         |
| Structural optimization viewer           | implementation/phase1/release/visualization/structural_optimization_viewer.html   mode=static |
| Optimized drawing review                 | implementation/phase1/release/visualization/optimized_drawing_review.html   projections=3   c |
| Authority routing warning                | none                                                                                          |
| Open gaps                                | P0=0 / P1=0 / P2=0                                                                            |
| MIDAS rows                               | total=12728, skipped=0, unknown=0                                                             |
| KDS rows                                 | compliance=511, member_checks=1056, clauses=16                                                |
| NDTHA residual                           | top=0.6880580090188355 m, drift=1.9136000000000006 %, fallback_rate=0.0                       |
| Registry artifacts                       | 12                                                                                            |
| Design-opt long feasible                 | True                                                                                          |
| Design-opt long final max DCR            | 0.9331636363636363                                                                            |
| Design-opt raw max drift                 | 9.200000000000003                                                                             |
| Design-opt repaired compliance max drift | 1.9872000000000005                                                                            |
| Design-opt compliance basis              | repaired_solver_validated_slice                                                               |
| Design-opt repair action count           | 1                                                                                             |

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|                                            |                                                                                               |
|--------------------------------------------|-----------------------------------------------------------------------------------------------|
| Design-opt constructability signal gain    | 0.6812506971740343                                                                            |
| Design-opt constructability avg            | 0.32310580388643245 -> 0.32090464334484636                                                    |
| Design-opt detailing complexity avg        | 0.4242652309769553 -> 0.4227255791394311                                                      |
| Design-opt selected family mix             | connection_detailing=5, detailing=6, perimeter_frame=1, rebar=2, slab_thickness=2, wall_thick |
| Design-opt selected dominant family        | wall_thickness (30.43%)                                                                       |
| Design-opt selected family mix trend       | connection_detailing=+0, detailing=+0, perimeter_frame=+0, rebar=+0, slab_thickness=+0, wa    |
| Design-opt previous dominant family        | wall_thickness (30.43%)                                                                       |
| Design-opt preview supply family mix       | beam_section=7, connection_detailing=14, detailing=142, perimeter_frame=1, rebar=254, slab    |
| Design-opt preview missing target families |                                                                                               |
| MGT export direct-patch changes            | 25                                                                                            |
| MGT export direct-patch families           | beam_section=1, connection_detailing=6, detailing=5, perimeter_frame=1, rebar=5, slab_thick   |
| MGT rebar namespace mode                   | group_local                                                                                   |
| MGT rebar delivery mode                    | direct_patch_eligible                                                                         |
| MGT evidence model                         | direct_patch_plus_zero_touch_verification_manifest                                            |
| MGT rebar material namespace present       | True                                                                                          |
| MGT rebar group-local namespace present    | True                                                                                          |
| MGT material rebar payloads                | 3/5                                                                                           |
| MGT group-local rebar payloads             | 6                                                                                             |

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|                                    |                                                                                        |
|------------------------------------|----------------------------------------------------------------------------------------|
| MGT rebar direct-patch eligible    | 6                                                                                      |
| MGT rebar direct-patch blockers    |                                                                                        |
| MGT rebar mapping sources          | alt_slab_wall_group_id=5, direct_group_id=1                                            |
| Design-opt cost delta              | 411.15925674073515                                                                     |
| Design-opt changed groups          | 20                                                                                     |
| Blocked cost-down rows             | 47                                                                                     |
| Blocked illegal-by-mask            | 0                                                                                      |
| Illegal-by-mask families           |                                                                                        |
| Blocked no-cost-gain               | 8                                                                                      |
| Blocked constructability hard-gate | 3                                                                                      |
| Hard-gate reasons                  | detailing_ratio_above_hard_limit=3                                                     |
| Hard-gate families                 | beam_section=1, rebar=2                                                                |
| No-cost-gain groups                | 5                                                                                      |
| No-cost-gain explain rows          | 5                                                                                      |
| Design-opt groups                  | Ablation=1/1   Dataset=1/1   Profile=1/1   Stage A=2/2   Stage A/B/C=1/1   Stage B=1/1 |
| Frame metrics                      | cases=3, drift_p95=4.167%, top_disp_p95=3.485%                                         |
| Wind metrics                       | cases=4, max_drift=0.000806%, residual_drift=0.000113%                                 |
| SSI metrics                        | cases=4, nonlinear_span=0.282342, residual_drift=0.004034%                             |

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Design-opt metrics

rows=13, raw\_drift=9.2000%, repaired\_drift=1.9872%, cost\_delta=411.159

This is the current external-validation submission baseline. Previous submission bundles were pruned after this package was created.

## Appendix: MIDAS Native Roundtrip / Write-Back Taxonomy

```
- 'roundtrip_gate_summary': `MIDAS native roundtrip: PASS | corpus=77 | native_text=35 | public_native=9 | public_raw_native=8 | public_bridge_native=1 | public_preview_native=7 | public_structural_preview_native=8 | fixture_native=4 | repo_native=3 | experiment_native=3 | archives=7 | ready=35 | public_ready=25 | public_native_ready=9 | public_raw_ready=8 | public_bridge_ready=1 | public_structural_preview_ready=8 | exact_queue=0/8 | korean_reconstruction=35 | korean_promotions=0/0 | fixture_ready=4 | repo_ready=3 | experiment_ready=3 | receipts=35/35 | topology=35/35 | load=35/35 | loadcomb=35/35 | exactI coverage=ready:100% | receipts:100% | loadcomb:100% | public_source:71% | exact_queue:100% | kr_recon:100% | korean_reconstruction:0 | korean_promotions:0 | taxonomy_exact:34 | canonical:1 | loss:0 | unsupported_manual:0 | pending_review=0 | public_source_ready=25 | public_bridge_ready=0 | public_raw_ready=8 | public_structural_preview_ready=7 | public_structural_preview_native=8 | taxonomy_exact:34 | canonical:1 | loss:0 | unsupported_manual:0 | pending_review=0`
- 'public_split': public_native_ready=9 | public_raw_ready=8 | public_bridge_ready=1 | public_archive_preview_ready=7 | public_structural_preview_ready=8 | public_source_ready=25 | fixture_ready=4 | repo_ready=3 | experiment_ready=3
- 'kr_preview_promoted': ready=8 | cand=8 | addl_now=0
```

```
- 'special_member_family': direct_patch='core_beam_connection_detailing=1, intermediate_beam_section=1, intermediate_wall_rebar=4, intermediate_wall_thickness=3,
perimeter_frame_column=1, perimeter_slab_thickness=2, perimeter_wall_detailing=5, perimeter_wall_rebar=1, perimeter_wall_thickness=2,
transfer_beam_connection_detailing=5' | supported='core_beam_connection_detailing=2, intermediate_beam_section=1, intermediate_wall_rebar=4,
intermediate_wall_thickness=3, perimeter_frame_column=1, perimeter_slab_thickness=2, perimeter_wall_detailing=10, perimeter_wall_rebar=1, perimeter_wall_thickness=2,
markdown='implementation/phase1/release/midas_native_roundtrip/exact_topology_structural_preview_promotion_queue.json' |
transfer_beam_connection_detailing=10' | zero_touch_verified='core_beam_connection_detailing=1, perimeter_wall_detailing=5, transfer_beam_connection_detailing=5'
- 'artifacts': gate_json='implementation/phase17/midas_native_roundtrip_gate_report.json' |
receipts_report='implementation/phase1/release/midas_native_roundtrip/midas_native_writeback_diff_receipts_report.json' |
appendix_md='implementation/phase1/release/midas_native_roundtrip/unsupported_lossy_card_family_appendix.md' |
appendix_json='implementation/phase1/release/midas_native_roundtrip/unsupported_lossy_card_family_appendix.json'
- 'promotion_queue_state': 'No supported exact topology archive candidates are pending right now. The queue reopens automatically when a new public archive decoded
preview lands with exact_topology_candidate=true.'
```

```
- taxonomy case counts: {"canonical_rewrite": 1, "lossy_rewrite": 0, "manual_review_required": 0, "parser_drop_suspected": 1, "preserved_exact": 34, "unsupported_card": 0}
```

```
- taxonomy_card_family_histogram: {"audit_manifest_action_families": {}, "audit_only_action_families": {}, "direct_patch_action_families": {"beam_section": 1, "connection_detailing": 6, "detailing": 5, "perimeter_frame": 1, "rebar": 5, "slab_thickness": 2, "wall_thickness": 5}, "supported_action_families": {"beam_section": 1, "connection_detailing": 12, "detailing": 10, "perimeter_frame": 1, "rebar": 5, "slab_thickness": 2, "wall_thickness": 5}, "unsupported_reason_counts": {}}
```

| Structure Type | Ready | Receipts | Topology | Load | LoadComb | Pending Review | Batch Markdown |

$$| \text{---} | \text{---} : | \text{---} : | \text{---} : | \text{---} : | \text{---} : | \text{---} : | \text{---} |$$

```
| beam | 2 | 2 | 2 | 2 | 2 | 0 | implementation/phase1/release/midas_native_roundtrip/beam.diff_batch.md |
```

```
| bearing | 1 | 1 | 1 | 1 | 1 | 0 | implementation/phase1/release/midas_native_roundtrip/bearing.diff batch.md |
```

```
| bridge | 2 | 2 | 2 | 2 | 2 | 0 | implementation/phase1/release/midas_native_roundtrip/bridge.diff_batch.md |
```

```
| bridge_section | 2 | 2 | 2 | 2 | 2 | 0 | implementation/phase1/release/midas_native_roundtrip/bridge_section.diff_batch.md |
```

```
| building | 11 | 11 | 11 | 11 | 11 | 0 | implementation/phase1/release/midas_native_roundtrip/building.diff_batch.md |
```

```
| foundation | 4 | 4 | 4 | 4 | 4 | 0 | implementation/phase1/release/midas_native_roundtrip/foundation.diff_batch.md |
```

```
| housing | 1 | 1 | 1 | 1 | 1 | 0 | implementation/phase1/release/midas_native_roundtrip/housing.diff_batch.md |
```

# Appendix: MIDAS Native Roundtrip / Write-Back Taxonomy

| industrial\_facility | 1 | 1 | 1 | 1 | 1 | 0 | implementation/phase1/release/midas\_native\_roundtrip/industrial\_facility.diff\_batch.md |

| mixed\_use | 2 | 2 | 2 | 2 | 2 | 0 | implementation/phase1/release/midas\_native\_roundtrip/mixed\_use.diff\_batch.md |

| public\_facility | 5 | 5 | 5 | 5 | 5 | 0 | implementation/phase1/release/midas\_native\_roundtrip/public\_facility.diff\_batch.md |

| ramp | 2 | 2 | 2 | 2 | 2 | 0 | implementation/phase1/release/midas\_native\_roundtrip/ramp.diff\_batch.md |

| stair | 2 | 2 | 2 | 2 | 2 | 0 | implementation/phase1/release/midas\_native\_roundtrip/stair.diff\_batch.md |

| Case | Structure Type | Mode | Pass | Pending Review | Summary | Receipt MD | Receipt JSON |

|---|---|---|---|---|---|---|

| gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback | bearing | public\_raw\_identity\_baseline | True | 0 | MIDAS native write-back diff receipt: PASS |  
case=gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback | type=bearing | topology=8/8 stable | load=5/5 stable | loadcomb=exact | direct\_patch=0 | pending\_review=0 |  
implementation/phase1/release/midas\_native\_roundtrip/01.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/02.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/03.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/04.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/05.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/06.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/07.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/08.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/09.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/10.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/11.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/12.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/13.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/14.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/15.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/16.gtc\_public\_bridge\_bearing\_c04\_\_identity\_writeback.midas\_native\_write-back\_diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/17.midas\_support\_stair\_archive\_structural\_preview\_native\_identity\_writeback.diff\_receipt.json |  
implementation/phase1/release/midas\_native\_roundtrip/18.ifc\_public\_award\_structure\_structural\_preview\_candidate\_identity\_writeback.diff\_receipt.md |  
implementation/phase1/release/midas\_native\_roundtrip/18.ifc\_public\_award\_structure\_structural\_preview\_candidate\_identity\_writeback.diff\_receipt.json |

[illegible]

## Appendix: Irregular Structure Track

- `track\_summary`: `Irregular structure collection gate: PASS | families=20 | sources=32 | local\_ready=15 | remote\_candidates=17 | collected=21 | top5=5`
- `benchmark\_manifest`: `implementation/phase1/release/external\_benchmark\_kickoff/irregular\_benchmark\_execution\_manifest.json`
- `counts`: `families=20 | sources=32 | local\_ready=15 | remote\_candidates=17 | collected=21 | native\_roundtrip\_candidates=18 | solver\_candidates=13 | ai\_candidates=32 | top5=0`
- `top5\_split`: `local\_ready=0 | remote\_needed=0 | family\_ids=n/a`
- `benchmark\_execution\_summary`: `Irregular benchmark execution: CHECK | ready=0 | blocked=0 | task\_count=0 | canonical=0 | bridged=0 | proxy=0 | receipts=0`

# Appendix: MIDAS KDS Row Provenance Export

MIDAS KDS row provenance export: PASS | combos=6 | rows=144 | members=12 | clauses=6 | exact\_rows=144 | clause\_filters=6 | member\_filters=12 | reverse\_jump=viewer\_subset\_reverse\_jump\_v11

json=implementation/phase1/release/kds\_compliance/midas\_kds\_row\_provenance\_table.json |  
csv=implementation/phase1/release/kds\_compliance/midas\_kds\_row\_provenance\_table.csv |  
report=implementation/phase1/release/kds\_compliance/midas\_kds\_row\_provenance\_table\_report.json

gLCB1 | member=MF-012 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member\_types=column

gLCB1 | member=MF-005 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-005 | case=C-TRN-005 | baseline=27441 | member\_types=column

gLCB1 | member=MF-012 | clause=KDS-INT-FRAME-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member\_types=column

gLCB1 | member=MF-010 | clause=KDS-MOMENT-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-001 | case=C-TST-001 | baseline=27441 | member\_types=column

gLCB1 | member=MF-012 | clause=KDS-SHEAR-Y-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member\_types=column

gLCB1 | member=MF-005 | clause=KDS-INT-FRAME-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-005 | case=C-TRN-005 | baseline=27441 | member\_types=column

# Appendix: MIDAS KDS Row Provenance Export

gLCB1 | member=MF-007 | clause=KDS-MOMENT-Y-001 | baseline=27425

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TRN-007 | case=C-TRN-007 | baseline=27425 | member\_types=wall

gLCB1 | member=MF-012 | clause=KDS-AXIAL-001 | baseline=27441

mode=exact row-level provenance | clause=rows=12 | members=12 | rules=1 | hazards=3 | inventory=review=C-TST-003 | case=C-TST-003 | baseline=27441 | member\_types=column

clause=KDS-AXIAL-001 | rows=24 | members=12 | combos=2 | top\_member=MF-012 | top\_dcr=1.065

clause=KDS-INT-FRAME-001 | rows=24 | members=12 | combos=2 | top\_member=MF-012 | top\_dcr=1.137

clause=KDS-MOMENT-Y-001 | rows=24 | members=12 | combos=2 | top\_member=MF-012 | top\_dcr=1.216

clause=KDS-MOMENT-Z-001 | rows=24 | members=12 | combos=2 | top\_member=MF-012 | top\_dcr=0.991

clause=KDS-SHEAR-Y-001 | rows=24 | members=12 | combos=2 | top\_member=MF-012 | top\_dcr=1.110

clause=KDS-SHEAR-Z-001 | rows=24 | members=12 | combos=2 | top\_member=MF-012 | top\_dcr=0.863

member=MF-001 | baseline=26878 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=MF-002 | baseline=27287 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=MF-003 | baseline=26878 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=MF-004 | baseline=27425 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=MF-005 | baseline=27441 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=MF-006 | baseline=27287 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

## Appendix: MIDAS KDS Row Provenance Export

member=MF-007 | baseline=27425 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

member=MF-008 | baseline=26878 | rows=12 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001

hazard=combined | rows=48 | members=4 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001 | top\_dcr=1.216

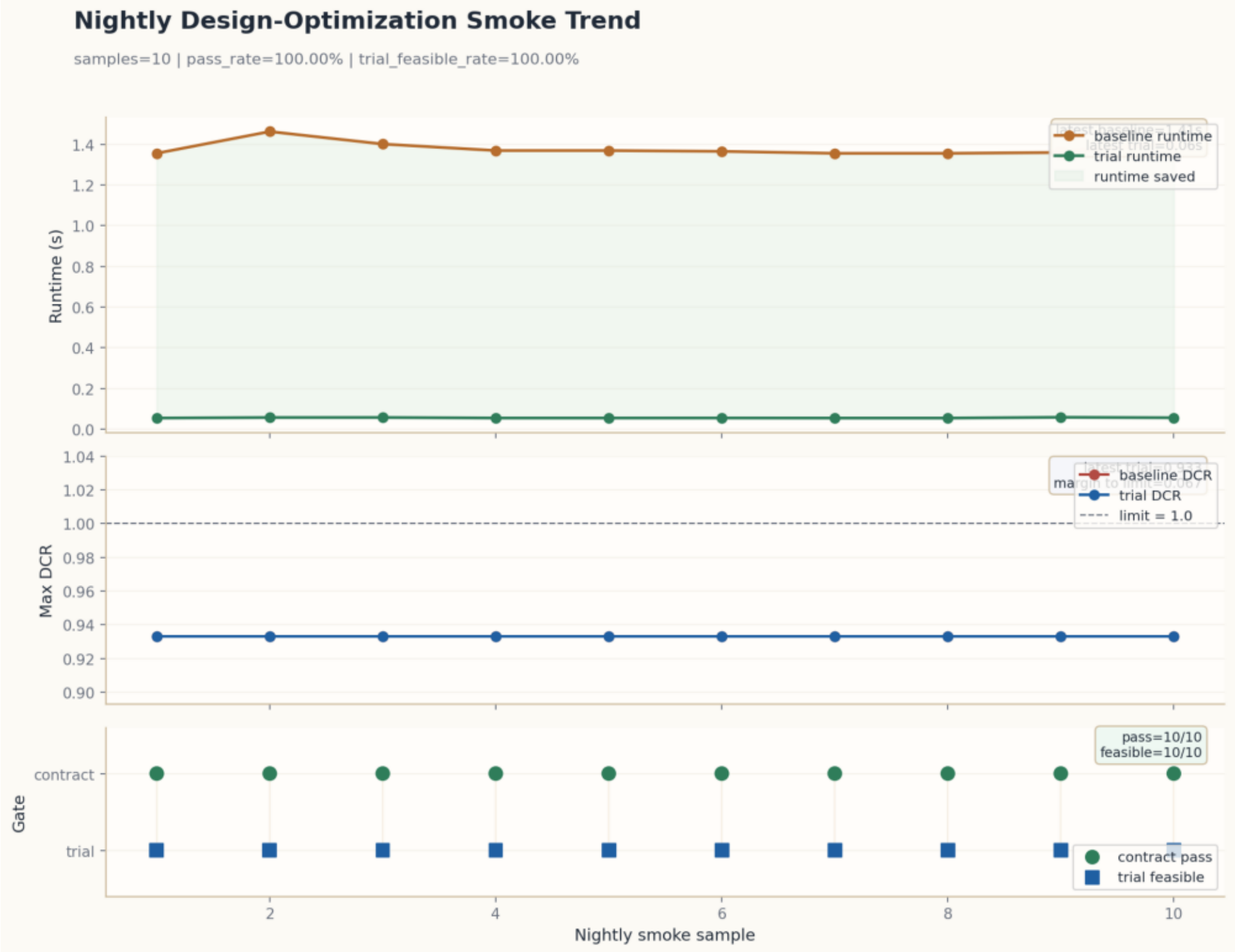
hazard=seismic | rows=48 | members=4 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001 | top\_dcr=1.112

hazard=wind | rows=48 | members=4 | clauses=6 | combos=2 | top\_clause=KDS-MOMENT-Y-001 | top\_dcr=0.809

rule\_family=strength | rows=120 | members=12 | hazards=3 | combos=2 | top\_clause=KDS-MOMENT-Y-001 | top\_dcr=1.216

rule\_family=strength\_interaction | rows=24 | members=12 | hazards=3 | combos=2 | top\_clause=KDS-INT-FRAME-001 | top\_dcr=1.137

# Nightly Smoke Trend



# Nightly Smoke Recent Samples

#6 2026-04-06T12:19:39.039411+00:00 | pass=True | trial\_feasible=True | baseline\_runtime=1.3646s | trial\_runtime=0.0554s | trial\_max\_dcr=0.9332 | action=connect

#7 2026-04-06T12:23:56.108705+00:00 | pass=True | trial\_feasible=True | baseline\_runtime=1.3550s | trial\_runtime=0.0550s | trial\_max\_dcr=0.9332 | action=connect

#8 2026-04-06T12:23:56.108705+00:00 | pass=True | trial\_feasible=True | baseline\_runtime=1.3550s | trial\_runtime=0.0550s | trial\_max\_dcr=0.9332 | action=connect

#9 2026-04-06T12:32:45.009996+00:00 | pass=True | trial\_feasible=True | baseline\_runtime=1.3590s | trial\_runtime=0.0590s | trial\_max\_dcr=0.9332 | action=connect

#10 2026-04-06T13:54:47.180033+00:00 | pass=True | trial\_feasible=True | baseline\_runtime=1.4059s | trial\_runtime=0.0567s | trial\_max\_dcr=0.9332 | action=connect

## Measured Chain Category Trend

